Notes on inverse mechanical design in `VertAX` using a stress-based target and centered EP

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1 Purpose of this note

These notes explain, in a self-contained way, how to formulate and implement an *inverse mechanical design* problem in `VertAX` using:

(1) a *stress-based* mechanical target (here, a shear modulus target),

(2) the existing `VertAX` architecture, which is currently built around minimizing a single inner energy on a single mesh,

(3) and a *second-order* (centered) Equilibrium Propagation (EP) gradient estimator, evaluated at the two nudged states $\pm\beta/2$.

The goal is to keep the implementation as close as possible to the current code structure: rather than rewriting `VertAX` around a joint state $\mathbf{Z} = (\mathbf{X}_+, \mathbf{X}_-)$, we use a *multi-condition wrapper* that repeatedly calls the same one-mesh inner solver.

The document is intentionally detailed. It is written for implementation, not only for presentation.

2 High-level idea

We want to design vertex-model parameters $\boldsymbol{\theta}$ such that the tissue reaches a prescribed *mechanical property*. The property considered here is the small-strain *static shear modulus* $G(\boldsymbol{\theta})$ of a periodic tissue.

The main point is that we do *not* define G from the curvature of the relaxed energy with respect to strain, because then the envelope theorem would remove much of the bilevel difficulty. Instead, we define G from the response of the *relaxed stress* to a small imposed shear. This makes the outer objective depend on the equilibrium configurations themselves, and therefore leads to a genuinely bilevel problem.

3 Step 1: Define stress from the energy by differentiating with respect to strain

Let $\mathcal{E}(\mathbf{X}, \boldsymbol{\theta})$ be the vertex-model energy, where:

- $\mathbf{X}$ denotes all vertex positions,
- $\boldsymbol{\theta}$ denotes all tunable model parameters (line tensions, target areas, shape factors, elastic moduli, *etc.*).

Let $\boldsymbol{\varepsilon} \in \mathbb{R}^{2\times 2}$ denote an infinitesimal homogeneous strain. We define deformed vertex positions by

$$\mathbf{x}_i(\boldsymbol{\varepsilon}) = (\mathbf{I} + \boldsymbol{\varepsilon})\mathbf{x}_i.$$

Equivalently, if we collect all vertices into $\mathbf{X}$, we write

$$\mathbf{X}(\boldsymbol{\varepsilon}) = (\mathbf{I} + \boldsymbol{\varepsilon})\mathbf{X}.$$

We then define a stress tensor from the derivative of the total energy with respect to strain:

$$\boldsymbol{\sigma}(\mathbf{X}, \boldsymbol{\theta}) = \frac{1}{A}\frac{\partial}{\partial \boldsymbol{\varepsilon}}\mathcal{E}(\mathbf{X}(\boldsymbol{\varepsilon}), \boldsymbol{\theta})\bigg|_{\boldsymbol{\varepsilon}=\mathbf{0}},$$

where A is the area of the periodic box.

Important practical point. In `VertAX`, this derivative can be computed with forward automatic differentiation (`jax.jacfwd`) with respect to the strain matrix $\boldsymbol{\varepsilon}$, *without* differentiating through the inner mechanical relaxation. This is much safer than backpropagating through the whole simulation.

Interpretation. For a given configuration $\mathbf{X}$, this produces a 2×2 tensor $\boldsymbol{\sigma}(\mathbf{X}, \boldsymbol{\theta})$. In particular, its off-diagonal component σ_{xy} measures the shear response of the current configuration.

4 Step 2: Define a shear modulus from *relaxed* stressed configurations

We now consider a small imposed shear $s \in \{+\gamma_0, -\gamma_0\}$, where $\gamma_0 > 0$ is small.

4.1 Choice of the imposed shear

We use the simple engineering-shear deformation gradient

$$\mathbf{F}(s) = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \qquad s \in \{+\gamma_0, -\gamma_0\}.$$

Applied to the vertices,

$$\mathbf{x}_i^{(s)} = \mathbf{F}(s)\,\mathbf{x}_i \quad \Longleftrightarrow \quad \begin{cases} x_i^{(s)} = x_i + s\,y_i, \\ y_i^{(s)} = y_i. \end{cases}$$

In a periodic system, keeping s fixed means that the geometry routines used to compute lengths, areas and perimeters must consistently interpret periodic offsets under the imposed shear. In practice, this means that the *same* scalar s must be passed to the PBC-aware geometric functions throughout the inner relaxation.

4.2 Strained inner problems

For each imposed shear $s \in \{+\gamma_0, -\gamma_0\}$, define a strained energy

$$\mathcal{E}_s(\mathbf{X}, \boldsymbol{\theta}),$$

which is the same vertex-model energy, but evaluated under the shear-dependent periodic geometry.

The corresponding equilibrium configuration is

$$\mathbf{X}_s^*(\boldsymbol{\theta}) \in \arg\min_{\mathbf{X}} \mathcal{E}_s(\mathbf{X}, \boldsymbol{\theta}).$$

Meaning. For a fixed parameter set $\boldsymbol{\theta}$:

- $\mathbf{X}^*_{+\gamma_0}(\boldsymbol{\theta})$ is the equilibrium tissue under a small positive shear,
- $\mathbf{X}^*_{-\gamma_0}(\boldsymbol{\theta})$ is the equilibrium tissue under a small negative shear.

4.3 Stress-based shear modulus

At each of these two equilibria, compute the shear stress:

$$\sigma_{xy}^{(+)}(\boldsymbol{\theta}) = \sigma_{xy}\big(\mathbf{X}^*_{+\gamma_0}(\boldsymbol{\theta}), \boldsymbol{\theta}\big)\,, \qquad \sigma_{xy}^{(-)}(\boldsymbol{\theta}) = \sigma_{xy}\big(\mathbf{X}^*_{-\gamma_0}(\boldsymbol{\theta}), \boldsymbol{\theta}\big)\,.$$

A natural finite-difference estimator of the static shear modulus is then

$$G(\boldsymbol{\theta}) = \frac{\sigma_{xy}^{(+)}(\boldsymbol{\theta}) - \sigma_{xy}^{(-)}(\boldsymbol{\theta})}{2\gamma_0}.$$

This quantity is genuinely bilevel because $\sigma_{xy}^{(\pm)}$ depends on the *equilibrium configurations* $\mathbf{X}^*_{\pm\gamma_0}(\boldsymbol{\theta})$, and not only on the minimized value of the same inner energy.

5 Step 3: Define the outer cost

Given a target modulus G_{tgt}, define

$$\mathcal{C}(\boldsymbol{\theta}) = \big(G(\boldsymbol{\theta}) - G_{\text{tgt}}\big)^2\,.$$

Expanding $G(\boldsymbol{\theta})$ gives

$$\mathcal{C}(\boldsymbol{\theta}) = \left(\frac{\sigma_{xy}^{(+)}(\boldsymbol{\theta}) - \sigma_{xy}^{(-)}(\boldsymbol{\theta})}{2\gamma_0} - G_{\text{tgt}}\right)^2.$$

This is the inverse mechanical design objective: tune $\boldsymbol{\theta}$ so that the tissue has the target shear modulus.

6 Step 4: Why this is a bilevel problem

The full optimization reads

$$\min_{\boldsymbol{\theta}} \mathcal{C}(\boldsymbol{\theta}) \quad \text{subject to} \quad \mathbf{X}^*_s(\boldsymbol{\theta}) \in \arg\min_{\mathbf{X}} \mathcal{E}_s(\mathbf{X}, \boldsymbol{\theta}), \quad s \in \{+\gamma_0, -\gamma_0\}.$$

There are therefore *two* inner equilibrium problems, one per imposed shear. Conceptually, this is equivalent to a single bilevel problem with a joint state

$$\mathbf{Z} = (\mathbf{X}_+, \mathbf{X}_-)$$

and total inner energy

$$\mathcal{E}_{\text{tot}}(\mathbf{Z}, \boldsymbol{\theta}) = \mathcal{E}_{+\gamma_0}(\mathbf{X}_+, \boldsymbol{\theta}) + \mathcal{E}_{-\gamma_0}(\mathbf{X}_-, \boldsymbol{\theta}),$$

but in practice `VertAX` is currently designed around one mesh and one inner energy. Therefore, we keep the architecture unchanged and solve the two inner problems with a wrapper.

7 Step 5: Adaptation strategy for current `VertAX` (Option 1: multi-condition wrapper)

7.1 Why a wrapper is enough

The current `VertAX` inner machinery already does:

(a) define one energy on one mesh,

(b) minimize it with respect to one set of vertex positions,

(c) return an equilibrium configuration.

To support the inverse mechanical design task above, we do *not* need to rewrite the whole code around a joint state. We only need a small wrapper that runs the same inner solver several times under different imposed conditions:

$$s = +\gamma_0, \qquad s = -\gamma_0.$$

7.2 Free (unperturbed) phase

For a given parameter set $\boldsymbol{\theta}$:

(1) build one problem instance with imposed shear $+\gamma_0$,

(2) relax it to obtain $\mathbf{X}^*_{+\gamma_0}(\boldsymbol{\theta})$,

(3) build another problem instance with imposed shear $-\gamma_0$,

(4) relax it to obtain $\mathbf{X}^*_{-\gamma_0}(\boldsymbol{\theta})$,

(5) compute the corresponding stress tensor in each case,

(6) compute $G(\boldsymbol{\theta})$ and then $\mathcal{C}(\boldsymbol{\theta})$.

7.3 Centered EP phase

EP requires two additional nudged evaluations, at $\beta = +\beta_0/2$ and $\beta = -\beta_0/2$.

Because the current code is one-mesh based, the easiest way to proceed is:

- treat the pair of strained meshes as a *joint state* only conceptually,
- but implement the nudged dynamics through repeated calls to the same one-mesh inner solver, while holding the other mesh fixed inside the current evaluation of the nudged cost.

This is simply a block-coordinate approximation of the joint nudged equilibrium.

8 Step 6: The centered EP construction

8.1 Joint state and total inner energy

Conceptually define

$$\mathbf{Z} = (\mathbf{X}_+, \mathbf{X}_-)$$

and

$$\mathcal{E}_{\text{tot}}(\mathbf{Z}, \boldsymbol{\theta}) = \mathcal{E}_{+\gamma_0}(\mathbf{X}_+, \boldsymbol{\theta}) + \mathcal{E}_{-\gamma_0}(\mathbf{X}_-, \boldsymbol{\theta}).$$

Define the stress-based modulus as a function of the joint state:

$$\widehat{G}(\mathbf{Z}, \boldsymbol{\theta}) = \frac{\sigma_{xy}(\mathbf{X}_+, \boldsymbol{\theta}) - \sigma_{xy}(\mathbf{X}_-, \boldsymbol{\theta})}{2\gamma_0}.$$

Then define the outer loss evaluated on a not-necessarily-relaxed pair of states:

$$\widehat{\mathcal{C}}(\mathbf{Z}, \boldsymbol{\theta}) = \left(\widehat{G}(\mathbf{Z}, \boldsymbol{\theta}) - G_{\text{tgt}}\right)^2.$$

8.2 Augmented EP energy

Introduce the augmented energy

$$\mathcal{L}(\mathbf{Z}, \boldsymbol{\theta}, \beta) = \mathcal{E}_{\text{tot}}(\mathbf{Z}, \boldsymbol{\theta}) + \beta\, \widehat{\mathcal{C}}(\mathbf{Z}, \boldsymbol{\theta}).$$

For $\beta = 0$, minimizing $\mathcal{L}$ is just minimizing the two strained inner energies independently.

For $\beta \neq 0$, the cost term couples the two states through the target modulus. This is why, in practice, we need alternating updates between the two meshes.

8.3 Free and nudged equilibria

Define:

$$\mathbf{Z}^*_{\boldsymbol{\theta},0} \approx \arg\min_{\mathbf{Z}} \mathcal{L}(\mathbf{Z}, \boldsymbol{\theta}, 0),$$
$$\mathbf{Z}^*_{\boldsymbol{\theta},+\beta/2} \approx \arg\min_{\mathbf{Z}} \mathcal{L}(\mathbf{Z}, \boldsymbol{\theta}, +\beta/2),$$
$$\mathbf{Z}^*_{\boldsymbol{\theta},-\beta/2} \approx \arg\min_{\mathbf{Z}} \mathcal{L}(\mathbf{Z}, \boldsymbol{\theta}, -\beta/2).$$

The centered EP estimator is then

$$\nabla_{\boldsymbol{\theta}} \mathcal{C}(\boldsymbol{\theta}) \approx \frac{1}{\beta} \left[\partial_{\boldsymbol{\theta}} \mathcal{L}(\mathbf{Z}^*_{\boldsymbol{\theta},+\beta/2}, \boldsymbol{\theta}, +\beta/2) - \partial_{\boldsymbol{\theta}} \mathcal{L}(\mathbf{Z}^*_{\boldsymbol{\theta},-\beta/2}, \boldsymbol{\theta}, -\beta/2) \right].$$

Why centered EP is preferred. The centered estimator is more accurate than the one-sided estimator and, empirically in `VertAX`, is the EP formulation that best matches AD and ID.

9 Step 7: How to compute the nudged equilibria using the existing one-mesh solver

9.1 The practical issue

The present `VertAX` code minimizes one energy with respect to one mesh:

$$\mathbf{X}^* \in \arg\min_{\mathbf{X}} \mathcal{E}(\mathbf{X}, \boldsymbol{\theta}).$$

It cannot directly minimize a joint energy with respect to two independent meshes simultaneously.

9.2 The workaround: alternating minimization

To approximate

$$\mathbf{Z}^*_{\boldsymbol{\theta},\beta} = (\mathbf{X}^*_{+,\beta}, \mathbf{X}^*_{-,\beta})$$

for a fixed value of β, proceed as follows.

Initialization. Start from the free equilibria:

$$\mathbf{X}^{(0)}_{+,\beta} = \mathbf{X}^*_{+\gamma_0}(\boldsymbol{\theta}), \qquad \mathbf{X}^{(0)}_{-,\beta} = \mathbf{X}^*_{-\gamma_0}(\boldsymbol{\theta}).$$

Alternating updates. At iteration k:

(1) update the $+$ state while keeping the $-$ state fixed:

$$\mathbf{X}^{(k+1)}_{+,\beta} \approx \arg\min_{\mathbf{X}_+} \left[\mathcal{E}_{+\gamma_0}(\mathbf{X}_+, \boldsymbol{\theta}) + \beta\, \widehat{\mathcal{C}}\big((\mathbf{X}_+, \mathbf{X}^{(k)}_{-,\beta}), \boldsymbol{\theta}\big) \right],$$

(2) update the $-$ state while keeping the $+$ state fixed:

$$\mathbf{X}^{(k+1)}_{-,\beta} \approx \arg\min_{\mathbf{X}_-} \left[\mathcal{E}_{-\gamma_0}(\mathbf{X}_-, \boldsymbol{\theta}) + \beta\, \widehat{\mathcal{C}}\big((\mathbf{X}^{(k+1)}_{+,\beta}, \mathbf{X}_-), \boldsymbol{\theta}\big) \right].$$

After a few alternating sweeps, one obtains an approximate nudged equilibrium pair.

Why this is acceptable. The aim of EP is not to solve an exact symbolic reformulation, but to estimate the outer gradient from nearby perturbed equilibria of the same simulator. Alternating relaxation is therefore perfectly consistent with the spirit of EP and with the current `VertAX` architecture.

10 Step 8: Explicit expression of $\partial_{\boldsymbol{\theta}}\mathcal{L}$

The centered EP estimator requires

$$\partial_{\boldsymbol{\theta}}\mathcal{L} = \partial_{\boldsymbol{\theta}}\mathcal{E}_{\text{tot}} + \beta\,\partial_{\boldsymbol{\theta}}\widehat{\mathcal{C}}.$$

Expand the first term:

$$\partial_{\boldsymbol{\theta}}\mathcal{E}_{\text{tot}} = \partial_{\boldsymbol{\theta}}\mathcal{E}_{+\gamma_0}(\mathbf{X}_+, \boldsymbol{\theta}) + \partial_{\boldsymbol{\theta}}\mathcal{E}_{-\gamma_0}(\mathbf{X}_-, \boldsymbol{\theta}).$$

Expand the second term:

$$\partial_{\boldsymbol{\theta}}\widehat{\mathcal{C}} = 2\left(\widehat{G}(\mathbf{Z}, \boldsymbol{\theta}) - G_{\text{tgt}}\right)\partial_{\boldsymbol{\theta}}\widehat{G}(\mathbf{Z}, \boldsymbol{\theta}),$$

with

$$\partial_{\boldsymbol{\theta}}\widehat{G}(\mathbf{Z}, \boldsymbol{\theta}) = \frac{\partial_{\boldsymbol{\theta}}\sigma_{xy}(\mathbf{X}_+, \boldsymbol{\theta}) - \partial_{\boldsymbol{\theta}}\sigma_{xy}(\mathbf{X}_-, \boldsymbol{\theta})}{2\gamma_0}.$$

Thus

$$\partial_{\boldsymbol{\theta}}\mathcal{L}(\mathbf{Z}, \boldsymbol{\theta}, \beta) = \partial_{\boldsymbol{\theta}}\mathcal{E}_{+\gamma_0} + \partial_{\boldsymbol{\theta}}\mathcal{E}_{-\gamma_0} + \beta\,2\left(\widehat{G} - G_{\text{tgt}}\right)\frac{\partial_{\boldsymbol{\theta}}\sigma_{xy}(\mathbf{X}_+, \boldsymbol{\theta}) - \partial_{\boldsymbol{\theta}}\sigma_{xy}(\mathbf{X}_-, \boldsymbol{\theta})}{2\gamma_0}.$$

Practical note. The stress itself is already computed by AD with respect to strain. Therefore, the parameter derivative of the stress can also be obtained by AD, but only through the *local* stress definition, not through the full relaxation process. This is a much shallower and safer derivative than unrolling the entire bilevel pipeline.

11 Step 9: Full algorithm

Inputs

- initial mesh and topology,
- model parameters $\boldsymbol{\theta}$,
- target shear modulus G_{tgt},
- small imposed shear amplitude γ_0,
- small EP perturbation amplitude β_0.

Outer iteration

For each outer iteration:

(1) **Free phase:**

  (a) relax the $+\gamma_0$ problem to obtain $\mathbf{X}^*_{+\gamma_0}(\boldsymbol{\theta})$,
  (b) relax the $-\gamma_0$ problem to obtain $\mathbf{X}^*_{-\gamma_0}(\boldsymbol{\theta})$,
  (c) compute $\sigma_{xy}^{(+)}$ and $\sigma_{xy}^{(-)}$,
  (d) compute $G(\boldsymbol{\theta})$ and $\mathcal{C}(\boldsymbol{\theta})$.

(2) **Positive nudge:** approximately minimize $\mathcal{L}(\mathbf{Z}, \boldsymbol{\theta}, +\beta_0/2)$ using alternating one-mesh relaxations, to obtain $\mathbf{Z}^*_{\boldsymbol{\theta},+\beta_0/2}$.

(3) **Negative nudge:** approximately minimize $\mathcal{L}(\mathbf{Z}, \boldsymbol{\theta}, -\beta_0/2)$ using alternating one-mesh relaxations, to obtain $\mathbf{Z}^*_{\boldsymbol{\theta},-\beta_0/2}$.

(4) **Gradient estimate:** compute

$$\widehat{\nabla_{\boldsymbol{\theta}}\mathcal{C}} = \frac{1}{\beta_0}\left[\partial_{\boldsymbol{\theta}}\mathcal{L}(\mathbf{Z}^*_{\boldsymbol{\theta},+\beta_0/2}, \boldsymbol{\theta}, +\beta_0/2) - \partial_{\boldsymbol{\theta}}\mathcal{L}(\mathbf{Z}^*_{\boldsymbol{\theta},-\beta_0/2}, \boldsymbol{\theta}, -\beta_0/2)\right].$$

(5) **Outer update:** update $\boldsymbol{\theta}$ with the chosen outer optimizer.

12 Step 10: Concrete code changes needed in `VertAX`

This section is implementation-oriented. It lists the code modifications required to build this pipeline on top of the current code base.

12.1 A. Add an imposed-shear parameter to geometry / energy evaluation

The current periodic geometry uses offsets stored in the half-edge table. To evaluate the model under imposed shear s, all geometric routines that use periodic offsets must be made shear-aware.

If the current periodic displacement uses

$$\Delta x = x_j - x_i + n_x L_x, \qquad \Delta y = y_j - y_i + n_y L_y,$$

replace it by the sheared version

$$\Delta x = x_j - x_i + n_x L_x + s\, n_y L_y, \qquad \Delta y = y_j - y_i + n_y L_y.$$

Code consequence. Functions such as:

- edge length,
- perimeter,
- area,
- any stress evaluation routine,

must receive one extra scalar argument, say `gamma_shear`, and use it whenever periodic offsets are converted to physical displacements.

12.2 B. Add a stress-from-strain helper

Create a helper that, given:

- a mesh at fixed state $\mathbf{X}$,
- parameters $\boldsymbol{\theta}$,
- a fixed imposed shear condition s,

returns the stress tensor

$$\boldsymbol{\sigma}(\mathbf{X}, \boldsymbol{\theta}) = \frac{1}{A}\frac{\partial}{\partial \boldsymbol{\varepsilon}}\mathcal{E}(\mathbf{X}(\boldsymbol{\varepsilon}), \boldsymbol{\theta})\bigg|_{\boldsymbol{\varepsilon}=0}.$$

Implementation idea:

- define a strain-dependent energy wrapper

$$\boldsymbol{\varepsilon} \mapsto \mathcal{E}(\mathbf{X}(\boldsymbol{\varepsilon}), \boldsymbol{\theta}),$$

- compute its Jacobian at zero strain with `jax.jacfwd`.

12.3 C. Add a one-condition relaxation wrapper

Create a high-level function of the form:

$$\texttt{relax_condition(mesh, theta, shear)} \to \mathbf{X}_s^*,$$

which:

(1) sets the imposed shear parameter,

(2) uses the existing `VertAX` inner solver on this one mesh,

(3) returns the relaxed configuration.

This function should not know anything about the outer objective.

12.4 D. Add a two-condition free evaluator

Create a function that:

(1) calls `relax_condition` with $+\gamma_0$,

(2) calls `relax_condition` with $-\gamma_0$,

(3) computes the two stresses,

(4) returns

$$\mathbf{X}_+^*, \quad \mathbf{X}_-^*, \quad \sigma_{xy}^{(+)}, \quad \sigma_{xy}^{(-)}, \quad G, \quad \mathcal{C}.$$

12.5 E. Add a two-condition nudged evaluator

Create a function that approximately minimizes

$$\mathcal{L}(\mathbf{Z}, \boldsymbol{\theta}, \beta) = \mathcal{E}_{+\gamma_0}(\mathbf{X}_+, \boldsymbol{\theta}) + \mathcal{E}_{-\gamma_0}(\mathbf{X}_-, \boldsymbol{\theta}) + \beta\,\widehat{\mathcal{C}}(\mathbf{X}_+, \mathbf{X}_-, \boldsymbol{\theta})$$

using alternating one-mesh relaxations.

Its inputs should include:

- initial guesses for $\mathbf{X}_+$ and $\mathbf{X}_-$ (typically the free equilibria),
- $\boldsymbol{\theta}$,
- the perturbation value β.

Its output should be the approximate nudged pair

$$(\mathbf{X}_{+,\beta}^*, \mathbf{X}_{-,\beta}^*).$$

12.6 F. Add a centered EP gradient wrapper

Create a function that:

(1) computes the free equilibria,

(2) computes the positively nudged equilibria at $+\beta_0/2$,

(3) computes the negatively nudged equilibria at $-\beta_0/2$,

(4) evaluates the centered EP estimator.

This wrapper is the mechanical-design analogue of the current EP machinery already present in `VertAX`, but now lifted to two sheared conditions.

13 Step 11: Recommended order of implementation

To reduce implementation risk, proceed in this order:

(1) Implement shear-aware geometry and verify that the inner solver works at fixed imposed shear.

(2) Implement stress computation from AD with respect to strain.

(3) Verify the stress numerically on known simple cases.

(4) Implement the free two-condition evaluator and check that $G(\boldsymbol{\theta})$ behaves sensibly as parameters vary.

(5) Implement the nudged two-condition evaluator for one-sided EP.

(6) Upgrade to centered EP at $\pm\beta_0/2$.

(7) Only then connect it to the outer optimizer.

14 Step 12: Practical caveats

- The imposed strain γ_0 should be small enough to remain in the linear regime, but large enough to avoid numerical cancellation in the stress difference.
- If T1 transitions are allowed under imposed shear, the estimated modulus may become path-dependent. For a first proof-of-concept, it may be preferable to choose γ_0 small enough that T1 events are rare or absent during the modulus measurement.
- When computing the stress by AD with respect to strain, it is safer to keep topology and offsets fixed during the strain derivative itself.
- The EP perturbation amplitude β_0 should be chosen so that nudged equilibria remain close to the free equilibria while producing a numerically stable difference.

15 Short implementation summary

The entire strategy can be summarized in one sentence:

> Keep the current `VertAX` one-mesh inner solver unchanged, run it under two opposite imposed shears, define a shear-modulus loss from the relaxed stresses, and estimate the outer gradient with centered EP by alternately nudging and relaxing the two conditions at $\pm\beta/2$.